How Rouge Works

The Name

In 1928, Ford opened the River Rouge Complex — a factory so vertically integrated that iron ore and rubber went in one end, and finished automobiles came out the other. Raw materials to finished product, under one roof.

Rouge is the software equivalent. A product idea goes in. A deployed, tested, monitored application comes out. No human writes code. No human triages bugs. The system builds, tests, evaluates, and ships — pausing only when it needs a product decision it cannot make itself.

Let's watch it build a product from scratch.

The Idea

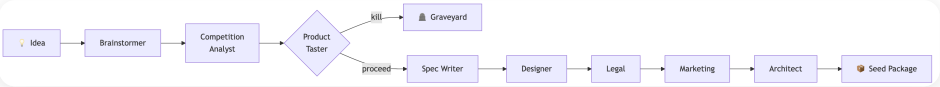
The product: **Epoch** — a focus timer that's beautiful enough to leave on screen.

A Pomodoro timer for knowledge workers who have tried every timer app and found them either ugly or bloated. The kind of app you leave visible on your second monitor all day — not because you have to, but because it's pleasant to look at.

- **Emotional north star**: "From 'where did the last 3 hours go?' to 'I can see my focus happening.'"
- **Aesthetic**: Techno-futuristic. Frosted glass, subtle glows, precise typography. Linear's design language meets a high-end watch face.
- **Scope**: Full Pomodoro cycle, configurable settings, session counter, keyboard shortcuts, synthesized audio chime. No accounts, no history beyond today, no themes, no task lists.
- **Differentiator**: Optimizes for *presence* — how it feels on your screen — not for features. The design IS the product.

Seeding: From Idea to Plan

The idea entered Rouge through **Rouge Spec**, the interactive seeding phase. A human describes what they want to build in a Slack conversation, and an eight-persona AI swarm turns that conversation into a buildable plan.



Each persona interrogates the idea from a different angle:

#	Persona	Role
1	**Brainstormer**	Expand the idea — 10-star version, emotional core, target user
2	**Competition Analyst**	Map the landscape — what exists, where's the gap
3	**Product Taster**	Challenge the idea — kill it or scope it
4	**Spec Writer**	Convert vision into feature areas + acceptance criteria
5	**Designer**	Three-pass: UX architecture → components → visual design
6	**Legal Advisor**	Privacy, terms, compliance
7	**Marketing Strategist**	Positioning, landing page, launch strategy
8	**Technical Architect**	Stack selection, infrastructure, deployment target

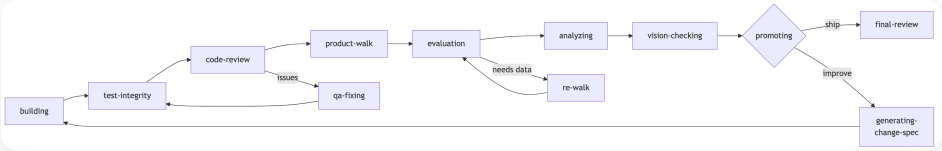
The human participates throughout — answering questions, making taste decisions, vetoing bad ideas. The Product Taster is the hard gate:

ideas that aren't worth building die before any specs are written.

The output is a **seed package**: vision document, six feature area specifications with 37 acceptance criteria, a complete design system, and a deployment target.

The Karpathy Loop: From Plan to Product

Rouge builds products through an iterative loop inspired by how neural networks train: build, evaluate, adjust, repeat. Each cycle advances the product toward convergence with the vision.



Phase	What happens	Browser?
building	Write code, run tests, deploy to staging	No
test-integrity	Verify all tests pass, no regressions	No
code-review	Engineering audit — architecture, dead code, security	No
product-walk	Open browser, screenshot every state, record observations	Yes
evaluation	Three lenses judge the observations (QA, Design, PO)	No
analyzing	Decide: ship, improve, or escalate	No
vision-checking	Compare built product against original vision	No
promoting	Advance to next cycle or trigger final review	No
final-review	Holistic "use it as a customer" walkthrough	Yes

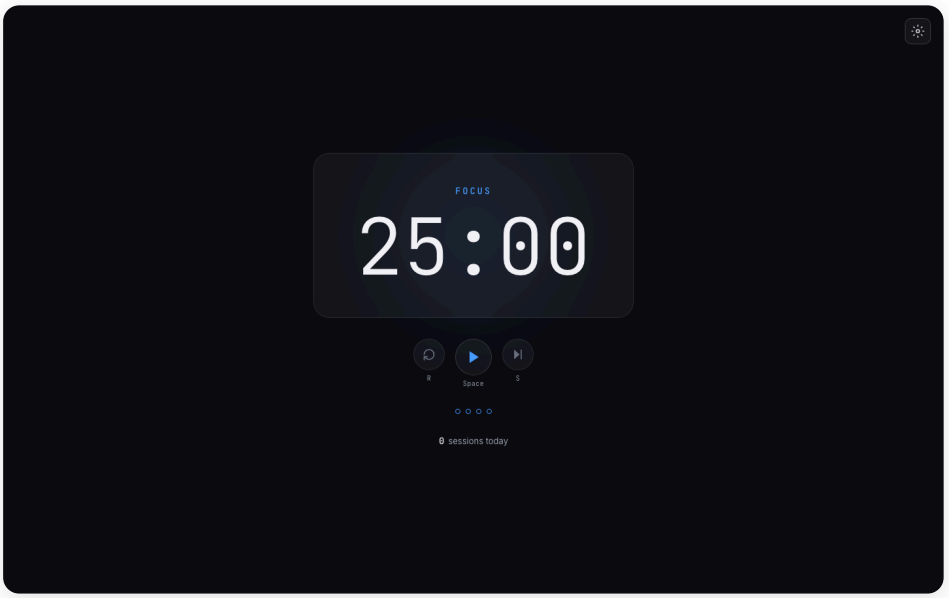
The browser opens exactly once per cycle. All three evaluation lenses judge the same observation data. This is the **observe-once, judge-through-lenses** architecture.

Building Epoch

Cycle 1: Everything at Once

Rouge built all six feature areas in a single cycle: Pomodoro engine, timer display, controls, cycle indicator, session counter, and settings modal. It chose Web Audio API for the chime (zero dependencies), CSS Modules for the frosted glass effects, and Date.now()-based timing (immune to background tab throttling).

89 tests passing. Deployed to Cloudflare Workers.



Cycle 2: The Quality Ratchet

The evaluation found concrete issues:

Issue	Fix
No crash recovery	ErrorBoundary with styled dark fallback
Settings modal used a div overlay	Native `<dialog>` element (automatic focus trap, Escape)
Footer text contrast 3.2:1	Fixed to WCAG AA compliance
No semantic landmarks	`<header>`, `<main>`, `<footer>`
Keyboard shortcuts undiscoverable	Inline hints below controls

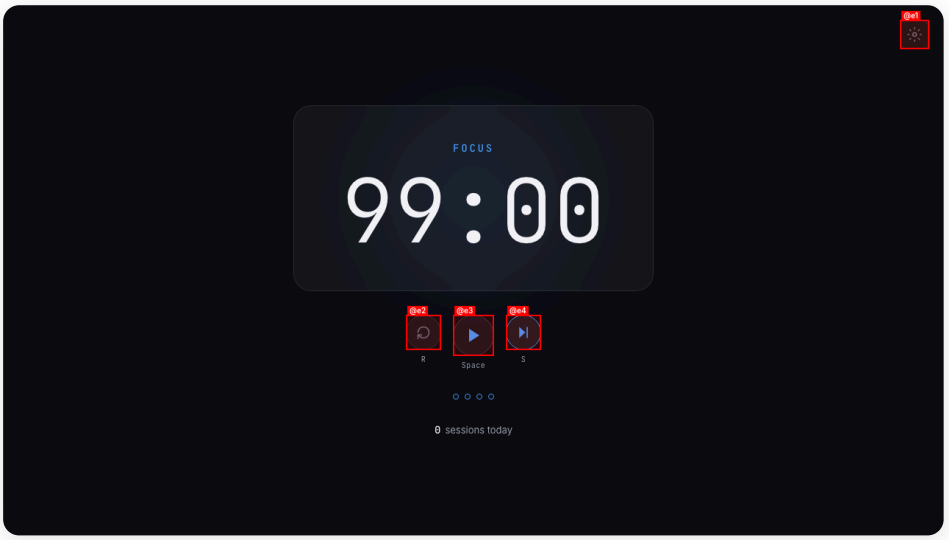
Health score: 68 → **82**. Accessibility: 89 → **100**.

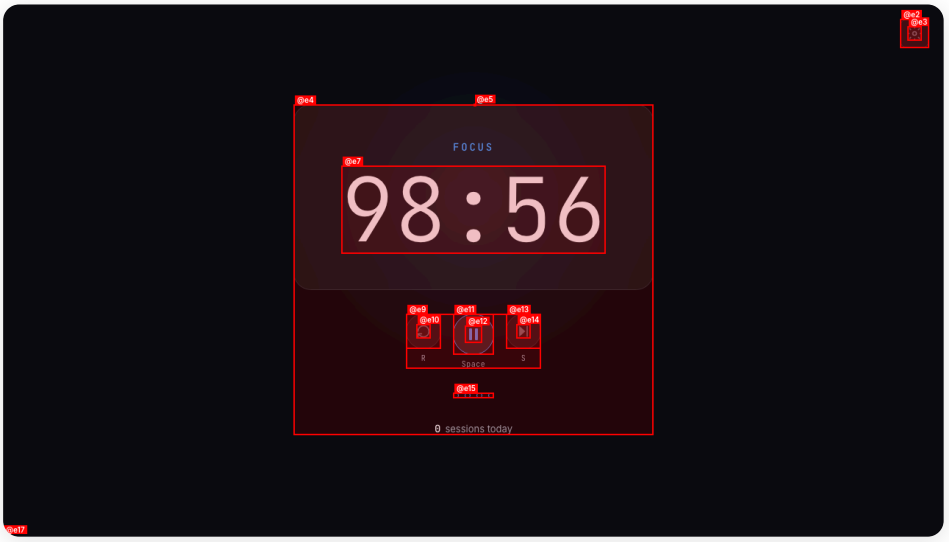
Cycles 3–5: Convergence

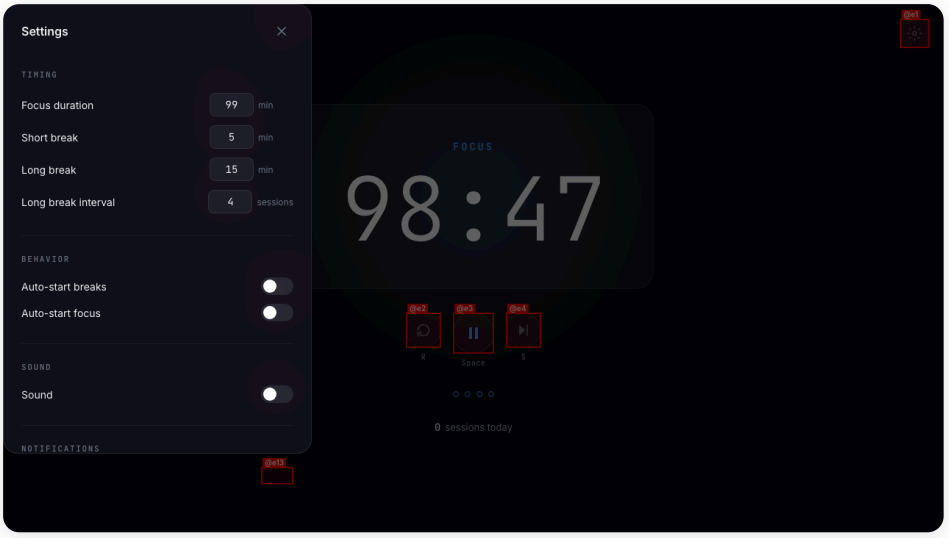
Each subsequent cycle refined further — polishing interactions, improving architecture, converging on the vision. The system tracked confidence across cycles and quality metrics stabilised.

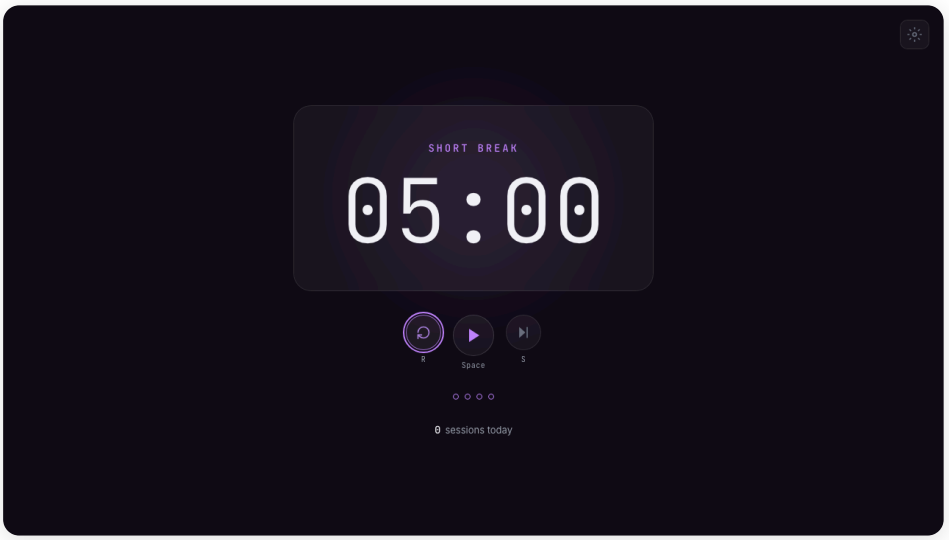
The Final Product

After 5 cycles, Rouge produced a deployed Pomodoro timer that met its vision.









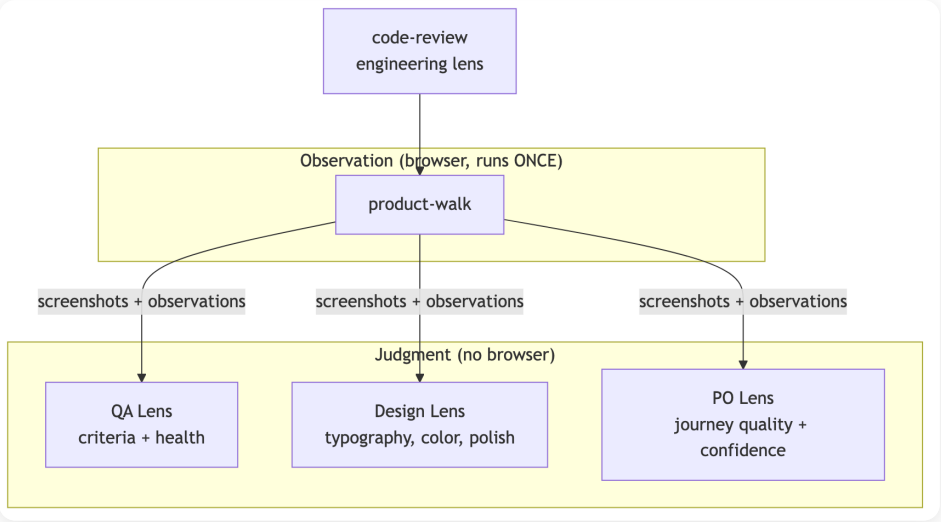
> "Epoch has converged strongly on its vision. The core promise — a focus timer beautiful enough to leave on screen — is delivered through a cohesive techno-futuristic aesthetic, atmospheric phase transitions, and purposeful minimalism."

Metric	Score
Tests passing	90
Acceptance criteria	67/67 (100%)
Health score	82/100
Lighthouse	91 / 100 / 100 / 100
AI code audit	90/100
Design score	86/100
AI slop score	6/100 (doesn't feel AI-generated)
PO confidence	0.89
Journey quality	All 6 rated "polished"

How It Judges

Rouge does not evaluate products through vibes. Every judgment is structured data with evidence.

Three lenses on one observation



code-review runs CLI tools: ESLint, dependency audit, duplication detection, dead code analysis, and a seven-dimension AI code audit. No browser needed.

product-walk opens the browser once, navigates every screen state, clicks every element, tests keyboard navigation, captures screenshots and accessibility trees. Then the browser closes.

evaluation reads both reports and applies three lenses:

- **QA lens**: Does the product match the spec? 67 criteria checked with evidence.
- **Design lens**: Does it look and feel like a real product? 8 categories scored, plus AI slop detection.
- **PO lens**: Will this delight customers? Journey quality, screen quality, confidence score.

Real tools, not vibes

Tool	Purpose
GStack	Headless Chromium — click, type, screenshot, test keyboard navigation
Lighthouse	Performance, accessibility, best practices, SEO
ESLint	Lint errors
jscpd	Duplication
madge	Circular dependencies
knip	Dead code
npm audit	Dependency vulnerabilities
AI code audit	Architecture, consistency, robustness, security, tech debt (7 dimensions)

Under the Hood

The stack

Rouge is a Node.js orchestrator that drives Claude via the Anthropic API. Every product it ships gets a full production stack, provisioned automatically:

Layer	Tool	Provisioned by
Hosting	Cloudflare Workers	Rouge — automatic
Database	Supabase (Postgres + Auth)	Rouge — automatic
Payments	Stripe	Rouge — sandbox, human activates production
Error monitoring	Sentry	Rouge — auto-creates project per product
Analytics	PostHog (EU)	Rouge — shared project, product-tagged
Session recording	PostHog	Included with analytics
CI/CD	GitHub Actions	Template — ships with project
Security headers	CSP, HSTS, X-Frame-Options	Template — ships with project
i18n	next-intl (Claude-translated)	Template — ships with project
Legal pages	Privacy + Terms scaffolds	Template — ships with project

Slack integration

Rouge is controlled entirely from Slack:

- `@Rouge` starts a seeding conversation
- App Home dashboard shows all projects with health scores, confidence, staging URLs
- `/rouge start`, `/rouge pause`, `/rouge ship`, `/rouge feedback`
- Phase transitions and escalations post as rich notifications
- Final review: human tests the staging URL at their own pace

Automatic management

- Deployment with health check and auto-rollback on failure
- Database migrations with dry-run preview and destructive operation detection
- State snapshots before every phase for corruption recovery
- Self-calibrating cost estimation from actual run data
- Cross-product learning: each completed product enriches a personal Library

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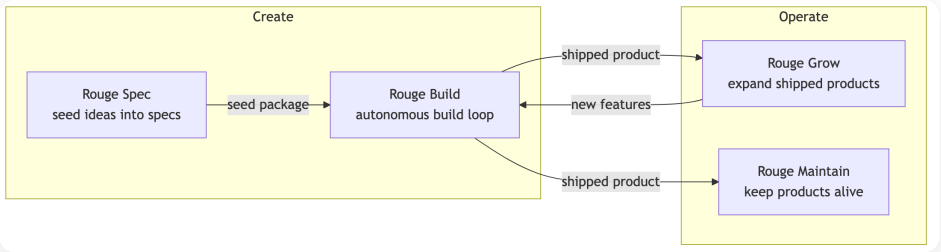
Cost

Epoch consumed **$8.53 of Opus compute** across 108 minutes of active execution. Five cycles, building through shipping.

A human developer would spend 1-2 weeks on the same scope. At $100-150/hour, that's $4,000-12,000. Rouge: under $10. Three orders of magnitude cheaper.

The real cost was building Rouge itself. But that amortizes across every product it builds after the first.

The Vision



Rouge Spec — "What to build"

Human + AI co-design a product through Slack conversation. Eight discipline-specific personas. Produces vision, specs, design, and infrastructure plan.

Rouge Build — "How to build it"

The Karpathy Loop. Autonomous build-evaluate-improve cycles until quality converges. This is what built Epoch.

Rouge Grow — "Make it better"

Feature expansion on shipped products. Reads analytics and user feedback. Builds new features while preserving existing user experience.

Rouge Maintain — "Keep it alive"

Autonomous upkeep. CVE scanning, bug triage, dependency updates, migration management. No new features — just keeping the foundation solid.

The goal: describe a product over coffee, approve a cost estimate, come back to a deployed, monitored, production-ready application.

Rouge handles everything between the idea and the first user.